

Intersectional Approaches for Supporting Casual Language and Culture Learning in Immigrant Families

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ABSTRACT

In multigenerational immigrant families, everyone is a lifelong learner. Grandparents must learn to foster social connection with their grandchildren despite language and culture barriers, while grandchildren seek to learn their heritage language and culture to better connect with their grandparents. Educational support tools for this context are sparse as language learning apps are often Eurocentric in design and do not fit in the existing routines of immigrant families. In my research, I design educational tools for marginalized immigration populations, such as apps for learning language, preserving family stories, and sharing culture. I design for people at the margins, which requires interdisciplinary intensive approaches. After providing an overview of my research, I present my ongoing project to build a storytelling tool for immigrant families called CrossRoads. CrossRoads employ human-centered approaches that are better suited for uncovering the needs and practices of marginalized immigrant populations. I design and evaluate a tool that fits within the routines of immigrant grandparents and grandchildren by employing the familiar activity of storytelling. I discuss preliminary findings, and implications of my work in the development of educational technology for supporting marginalized, lifelong learners in casual contexts.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing.

KEYWORDS

Learning, educational technology, creativity, immigrant, marginalized, families

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1 RESEARCH OVERVIEW

The potential to learn is in everything we do. Though most educational technology research focuses on classroom-centered contexts,

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a great bulk of learning takes place in the informal, non-structured routines of daily life. Prominently, first language acquisition generally takes place within homes and communities. However, this is untrue for immigrant families who cannot easily preserve heritage language and culture knowledge in their new society. Learning language and culture in the wild, in a society that is not designed to support such learning, is challenging because learning opportunities are sporadic, unstructured, and insufficient. These obstacles result in high rates of heritage language and culture loss among immigrant children, leading to adverse consequences such as social disconnect and familial conflict [5].

My research supports lifelong learning at points of disruption, such as migration. In my current PhD project, I collaborated with immigrant families to identify their needs for a language and culture support tool [12], and am now evaluating the resulting tool that mediates this exchange through storytelling. In my Master's work, I worked with adult English Language Learners (ELLs) searching for employment and built a tool that provided timely feedback on writing. These ELLs were not in school, and thus had no access to writing feedback. The tool leveraged peer and automated feedback to support writing needs and practices [16, 17]. I studied the sociocultural aspects of AI-based support by investigating learner perceptions of peer and automated feedback [11].

I also develop novel frameworks, human-centered design approaches, and mixed-methods that challenge historical technodeterminist approaches to designing for marginalized populations. I have reported on how employing Participatory Design and Contextual Inquiry in educational research makes learner decision-making processes explicit [13], suggested a mixed-methods, inductive-deductive approach to grounded theory [15], and offered methodological considerations for community engagement [14]. To achieve these multifaceted goals, I cross disciplinary domains (computer science, education, psychology), and contexts of use, by collaborating with academic and community-based partners. Thus, I have two overarching research questions that encompass two, interconnected dimensions:

1. How can we develop mixed-methods approaches for respectfully and ethically engaging with historically marginalized communities?
2. How can we reimagine the role of technology in fostering creative learning experiences?

To illustrate my research, I introduce one of my ongoing projects called *CrossRoads*.

2 WORK IN PROGRESS

The timeless social activity of passing down oral stories preserves family memory, identity, values, and culture [18]. However, this connection, particularly crucial across grandparent and grandchild

generations, is threatened by the language and cultural barriers emerging from displacement and migration [22]. As a result, immigrant grandparents and their young grandchildren struggle with fostering social kinship, leading to social disconnect and loss of cultural heritage [25]. Supporting language and culture learning in contexts that are not specialized for learning (i.e. daily family life) is challenging. This is evidenced by the fact that numerous studies report that though immigrant families typically express a strong desire to pass down heritage culture and language to their children, they are unable to do so [2, 5, 20–22, 25].

The unique contexts faces by immigrant families require flexible tech that makes popular learning apps like Duolingo inadequate for several major reasons. These apps do not fit into the existing routines of families, as grandparents and grandchildren typically do not engage with one another through pre-scheduled, structured lesson plans. The content of their courses lack flexibility as they may not align with the knowledge grandparents are interested in imparting. They also cannot teach tacit, cultural knowledge. There is a large overhead in creating content for each language, leading to the omission of certain languages. For instance, Duolingo does not have content for learning Urdu or Persian [7] despite being the heritage language of many immigrants [10], but has resources for the fictional Klingon language. These omissions are indicative of language learning apps' Eurocentric design process, where resources are developed based on the preferences and suggestions of adult English language speakers [9]. Therefore, these mobile learning apps are not suitable for long-term, sustained use by children and their grandparents. However, not enough is known about the needs and practices of these learners to design new technology for them.

Additionally, these apps are not intended as a collaborative, two-way learning experience. In immigrant families, language and culture exchange is a mechanism for fostering social connection between grandparents and grandchildren [12]. Tools designed for families should have both a learning focus and a collaboration focus as collaboration enriches the learning process by providing opportunities for shared sense-making. There is also a social process involved in collaboration, where peers must learn to articulate ideas, negotiate disagreements and reach shared resolutions [3]. Thus, collaboration presents opportunities not just for learning, but for fostering social connection between learners, in this case, grandparents and grandchildren [4]. Thus, there is a clear need to understand the needs and practices of immigrant grandparents and grandchildren, who have historically been underserved and overlooked in the technology design process.

2.1 Overview of CrossRoads

Our studies consist of an on-going series of projects in two phases. In the first, completed phase, we engaged with immigrant families across Canada in ethnographic Contextual Inquiry (CI) and Participatory design (PD) activities to uncover their design needs for a storytelling tool. We center our study on the practice of storytelling, which is an activity that is familiar to families [13]. PD integrates users into the technology creation process through a variety of methods (e.g., interviews, observations, or design activities) to elicit requirements from the early stages of the design

process [14]. Contextual inquiry (CI), an observational method, allows users to demonstrate their processes in their natural setting [18]. Together, these two techniques can reveal hidden insights into learner motivation and behaviour. We met with grandparents and grandchildren (ages 7–13) to understand how they share stories, culture, and language. Families were also involved in developing design ideas for a support tools. Grounded in this collaboration, we identified several design implications. We built and are in the process of evaluating the resulting tool, *StoryTapestry*, in the second phase.

2.1.1 The StoryTapestry App. StoryTapestry is a web and tablet based app to support reminiscence in immigrant families. The app guides co-located grandparent and grandchild dyads through the storytelling and story-making process. Families are first presented with a series of prompts to support the grandparent in recalling a memory. After sharing a story, which is audio-recorded, the dyad works together to create a digital storybook artifact to visually represent the story. The tool contains a collection of culturally relevant and diverse “stickers” that families drag-and-drop to create personalized artwork. Through this collaborative artifact creation process, families engage in discussion, negotiation, and co-regulation.

2.1.2 Methodology. In the first, completed phase, we employed CI and PD to understand the process of cultural exchange by introducing physical artifacts into the oral storytelling tradition. We achieved this by asking families to share stories. Families then drew out their stories on index cards, sticky notes, and posterboard, and worked together to create a system for organizing their memories. Ten workshops with grandparent-grandchild dyads from immigrant families were conducted in locations across Canada. These locations range from small rural communities to large metropolitan areas. Thematic analysis was conducted to analyze the sessions using an inductive coding process proposed by Braun and Clarke [3]. From the thematic analysis, we proposed several design guidelines. For a detailed overview of the first phase, see [12]. In the ongoing second phase, grandparent-grandchild dyads interact with the app by telling a story and creating an artifact to represent the story. In our analysis, we will validate the design guidelines from the first phase.

2.2 Preliminary Findings

Through this collaboration with multi-generational and culturally-at-risk immigrant families through human-centered design activities, we uncovered key insights into the learning and collaboration process between immigrant grandparents and grandchildren. For example, we observed in the ethnographic sessions that grandparents struggle to remember old memories, especially from their childhood. To support their reminiscence, the tool first presents a series of prompts to initiate conversation between grandchild and grandparent. This step then naturally snowballs into more dynamic conversation and creative sense-making between the generations. Breaking down the steps of a larger complex process in this order created a more enjoyable storytelling experience for families and produced richer audio and visual artifacts. We also identified opportunities for two-way learning by observing that children will

take on leadership roles when given the space for it. Thus, our storytelling tool is centered around visual artifact creation, as children feel more comfortable guiding their grandparents through such activities. As a result, an egalitarian collaboration is established with grandparents and grandchildren alternating between learner and educator roles.

3 EXPECTED CONTRIBUTIONS OF RESEARCH

Society has long held a misguided belief that creative activity is the domain of a select few “talented” individuals. However, many of us engage in creative acts in our work and hobbies. An impact I aim to make in my overall research is to demonstrate that through well designed tools, creativity can be fostered and learned. To demonstrate this, I work with immigrant families to design tools for informal culture and language learning.

For scientific and technical contributions, my goal is to re-imagine the role of AI-based support. There are gaps in the support available for learning creative skills in real world contexts, because research generally has not focused on how messy learning tasks can be supported. AI tools have been developed for procedural tasks such as solving math equations [23, 24], but less focus has been given to complex, open-ended tasks with no single right solution. Leveraging techniques such as humans-in-the-loop and adaptive prompting, I aim to build and test tools that augment a users’ creative learning experiences.

Another major research goal is the promotion of data science for social good. I am an advocate of ethical technology and a lot of my prior work has focused on the ethics of research and why technodeterminist approaches fail in the real world. I collaborate with my users at every stage of the creation process through novel mixed-methods, humans-in-the loop approaches. Through a human first approach, AI becomes a tool for augmenting user experience, but only if the users themselves reveal a place for it [8]. I have published on how my multilayered approaches makes decision-making processes explicit to design better tools, generates novel insights, and increases tool adoption in ways that traditional methods cannot [13].

REFERENCES

- [1] Behnke, A.O., Taylor, B.A. and Parra-Cardona, J.R. 2008. “I Hardly Understand English, But...”: Mexican Origin Fathers Describe Their Commitment as Fathers Despite the Challenges of Immigration. *Journal of Comparative Family Studies*. 39, 2 (2008), 187–205.
- [2] Braun, V. and Clarke, V. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology*. 3, 2 (Jan. 2006), 77–101. DOI:https://doi.org/10.1191/1478088706qp0630a.
- [3] Chi, M.T.H. and Wylie, R. 2014. The ICAP Framework: Linking Cognitive Engagement to Active Learning Outcomes. *Educational Psychologist*. 49, 4 (Oct. 2014), 219–243. DOI:https://doi.org/10.1080/00461520.2014.965823.
- [4] Choi, Y., He, M. and Harachi, T.W. 2008. Intergenerational Cultural Dissonance, Parent–Child Conflict and Bonding, and Youth Problem Behaviors among Vietnamese and Cambodian Immigrant Families. *Journal of Youth and Adolescence*. 37, 1 (2008), 85–96. DOI:https://doi.org/10.1007/s10964-007-9217-z.
- [5] Free Language Courses for English Speakers - Duolingo: https://www.duolingo.com/courses. Accessed: 2021-05-14.
- [6] Google PAIR 2019. *People + AI Guidebook*.
- [7] How can I suggest a new language course? – Duolingo Help Center: https://support.duolingo.com/hc/en-us/articles/204979660-How-can-I-suggest-a-new-language-course-. Accessed: 2021-05-14.
- [8] Immigration and Ethnocultural Diversity Highlight Tables - Immigrant population by place of birth, period of immigration, 2016 counts, both sexes, age (total), Canada, 2016 Census – 25% Sample data: 2017. https://www12.statcan.gc.ca/census-recensement/2016/dp-pd/hltfst/imm/Table.cfm?Lang=E&T=21&Geo=01. Accessed: 2020-06-01.
- [9] Liaquat, A., Akcayir, G., Epp, C.D. and Munteanu, C. 2019. Mature ELLs’ Perceptions Towards Automated and Peer Writing Feedback. *Transforming Learning with Meaningful Technologies* (Sep. 2019), 266–279.
- [10] Liaquat, A., Axtell, B. and Munteanu, C. 2021. Participatory Design for Intergenerational Culture Exchange in Immigrant Families: How Collaborative Narration and Creation Fosters Democratic Engagement. *Proceedings of the ACM on Human-Computer Interaction*. 5, CSCW1 (Apr. 2021), 1–40. DOI:https://doi.org/10.1145/3449172.
- [11] Liaquat, A., Axtell, B., Munteanu, C. and Demmans Epp, C. 2018. Contextual Inquiry, Participatory Design, and Learning Analytics: An Example. *Companion Proceedings of the 8th International Conference on Learning Analytics & Knowledge* (Sydney, Australia, 2018).
- [12] Liaquat, A. and Munteanu, C. 2021. Hidden Margins: Reflections on Designing with Populations at the Intersection. In *Companion of the 2019 ACM Conference on Human Factors in Computing Systems (CHI '19)* (Yokohama, Japan, May 2021).
- [13] Liaquat, A. and Munteanu, C. 2019. Social Learning Frameworks for Analyzing Collaboration with Marginalized Learners. *Conference Companion Publication of the 2019 on Computer Supported Cooperative Work and Social Computing* (Austin TX USA, Nov. 2019), 288–292.
- [14] Liaquat, A. and Munteanu, C. 2018. Towards a Writing Analytics Framework for Adult English Language Learners. In *Proceedings of the Seventh International Learning Analytics & Knowledge Conference (LAK '17)* (Sydney, Australia, 2018).
- [15] Liaquat, A., Munteanu, C. and Demmans Epp, C. 2020. Collaborating with Mature English Language Learners to Combine Peer and Automated Feedback: a User-Centered Approach to Designing Writing Support. *International Journal of Artificial Intelligence in Education*. (Jul. 2020). DOI:https://doi.org/10.1007/s40593-020-00204-4.
- [16] Lindley, S.E. 2012. Before I forget: From personal memory to family history. *Human-Computer Interaction*. 27, 1–2 (2012), 13–36.
- [17] Muller, M.J. and Kuhn, S. 1993. Participatory Design. *Commun. ACM*. 36, 6 (Jun. 1993), 24–28. DOI:https://doi.org/10.1145/153571.255960.
- [18] Nauck, B. 2001. Intercultural Contact and Intergenerational Transmission in Immigrant Families, Intercultural Contact and Intergenerational Transmission in Immigrant Families. *Journal of Cross-Cultural Psychology*. 32, 2 (Mar. 2001), 159–173. DOI:https://doi.org/10.1177/0022022101032002004.
- [19] Park, S.M. and Sarkar, M. 2008. Parents’ Attitudes Toward Heritage Language Maintenance for Their Children and Their Efforts to Help Their Children Maintain the Heritage Language: A Case Study of Korean-Canadian Immigrants. *Language, Culture and Curriculum*. (Dec. 2008). DOI:https://doi.org/10.2167/lcc337.0.
- [20] Treas, J. and Mazumdar, S. 2004. Kinkeeping and Caregiving: Contributions of Older People in Immigrant Families*. *Journal of Comparative Family Studies; Calgary*. 35, 1 (Winter 2004), 105–122.
- [21] Williams, J.J., Kim, J., Rafferty, A., Maldonado, S., Gajos, K.Z., Lasecki, W.S. and Heffernan, N. 2016. AXIS: Generating Explanations at Scale with Learnersourcing and Machine Learning. *Proceedings of the Third (2016) ACM Conference on Learning @ Scale* (New York, NY, USA, 2016), 379–388.
- [22] Williams, J.J., Rafferty, A.N., Tingley, D., Ang, A., Lasecki, W.S. and Kim, J. 2018. Enhancing Online Problems Through Instructor-Centered Tools for Randomized Experiments. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems - CHI '18* (Montreal QC, Canada, 2018), 1–12.
- [23] Wixon, D., Holtzblatt, K. and Knox, S. 1990. *Contextual design: an emergent view of system design*.
- [24] Xie, X. and Xia, Y. 2011. Grandparenting in Chinese Immigrant Families. *Marriage & Family Review*. 47, 6 (Sep. 2011), 383–396. DOI:https://doi.org/10.1080/01494929.2011.594218.